Lecture 4 Numerically solve RBC model. (23)

9 Solve LOM.

$$\begin{cases} \hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\gamma}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)\hat{c}_t \\ \sigma E_t \hat{c}_{t+1} + \tilde{\gamma}(1-\alpha)\hat{k}_t - \tilde{\gamma} E_t \hat{z}_{t+1} = \sigma \hat{c}_t \\ \hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t \end{cases}$$

State variables $\hat{k}_{t-1}$ and a shock $\hat{z}_t$ (pre-determined)

The dynamics of the model can be described as recursive law of motion in terms of states.

$$\text{LOM} \begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t \end{cases}$$

Solve for $(V_{kk}, V_{kz}, V_{ck}, V_{cz})$
待定系数法

① $\hat{z}_{t+1} = \psi\hat{z}_t + \varepsilon_t \Rightarrow E_t\hat{z}_{t+1} = \psi\hat{z}_t$

Since $E_t(\varepsilon_t) = 0$, $\varepsilon_t \sim iid\ N(0, \sigma^2)$

$E_t(\hat{z}_t) = \hat{z}_t$, z_t is already realized in t, thus in t's information set.

② $\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\gamma}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)\hat{c}_t \Rightarrow V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t = \frac{1}{\beta}\hat{k}_{t-1} - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)$
$(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t) + \frac{\tilde{\gamma}}{\alpha\beta}\hat{z}_t \Rightarrow \left[\frac{1}{\beta} - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)V_{ck} - V_{kk}\right]\hat{k}_{t-1} +$
$\left[\frac{\tilde{\gamma}}{\alpha\beta} - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)V_{cz} - V_{kz}\right]\hat{z}_t = 0$

$$\Rightarrow \begin{cases} V_{kk} = \frac{1}{\beta} - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)V_{ck} \quad \cdots (1) \\ V_{kz} = \frac{\tilde{\gamma}}{\alpha\beta} - \left(\frac{\tilde{\gamma}}{\alpha\beta} - \delta\right)V_{cz} \quad \cdots (2) \end{cases}$$

③ $\sigma E_t\hat{c}_{t+1} + \tilde{\gamma}(1-\alpha)\hat{k}_t - \tilde{\gamma}E_t\hat{z}_{t+1} = \sigma\hat{c}_t \Rightarrow \sigma E_t[V_{ck}\hat{k}_t + V_{cz}\hat{z}_{t+1}] + \tilde{\gamma}(1-\alpha)$
$(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t) - \tilde{\gamma}\psi\hat{z}_t = \sigma(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t)$

$\Rightarrow \sigma E_t[V_{ck}(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t) + V_{cz}\hat{z}_{t+1}] + \tilde{\gamma}(1-\alpha)V_{kk}\hat{k}_{t-1} + \tilde{\gamma}(1-\alpha)V_{kz}\hat{z}_t$
$- \tilde{\gamma}\psi\hat{z}_t = \sigma V_{ck}\hat{k}_{t-1} + \sigma V_{cz}\hat{z}_t$

$\Rightarrow [\sigma V_{ck}V_{kk} + \tilde{\gamma}(1-\alpha)V_{kk} - \sigma V_{ck}]\hat{k}_{t-1} + [\sigma(V_{ck}V_{kz} + V_{cz}\psi) + \tilde{\gamma}(1-\alpha)V_{kz}$
$- \tilde{\gamma}\psi - \sigma V_{cz}]\hat{z}_t = 0$

$$\Rightarrow \begin{cases} \sigma V_{ck}(1 - V_{kk}) = \tilde{\gamma}(1-\alpha)V_{kk} \quad \cdots (3) \\ \sigma V_{cz}(1-\psi) = [\sigma V_{ck} + \tilde{\gamma}(1-\alpha)]V_{kz} - \tilde{\gamma}\psi \quad \cdots (4) \end{cases}$$

Collect 4 equations, solve for 4 unknown:

(i) (1) & (3) are two equations about (V_{kk}, V_{ck}), solve for V_{kk} by substituting V_{ck} out.

$$\sigma\left(\frac{1}{\beta}-V_{kk}\right)\cdot\frac{1}{\frac{\tilde{\gamma}}{\alpha\beta}-\delta}(1-V_{kk}) = \tilde{\gamma}(1-\alpha)V_{kk}$$

$$\Rightarrow \frac{\sigma}{\beta}-\frac{\sigma}{\beta}V_{kk}-\sigma V_{kk}+\sigma V_{kk}^2 = \tilde{\gamma}(1-\alpha)\left(\frac{\tilde{\gamma}}{\alpha\beta}-\delta\right)V_{kk}$$

$$\Rightarrow V_{kk}^2-\frac{1}{\beta}V_{kk}-V_{kk}-\frac{\tilde{\gamma}(1-\alpha)\left(\frac{\tilde{\gamma}}{\alpha\beta}-\delta\right)}{\sigma}V_{kk}+\frac{1}{\beta}=0$$

$$\Rightarrow V_{kk}^2-\left[1+\frac{1}{\beta}+\frac{\tilde{\gamma}(1-\alpha)\left(\frac{\tilde{\gamma}}{\alpha\beta}-\delta\right)}{\sigma}\right]V_{kk}+\frac{1}{\beta}=0$$

Let $\gamma = 1+\frac{1}{\beta}+\frac{\tilde{\gamma}(1-\alpha)}{\sigma}\left(\frac{\tilde{\gamma}}{\alpha\beta}-\delta\right)$

$$V_{kk,1} = \frac{\gamma-\sqrt{\gamma^2-\frac{4}{\beta}}}{2} = \frac{\gamma}{2}-\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}} > 0$$

$$V_{kk,2} = \frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}} > 0$$

We need to choose stable root out of the linear rational expertation (LRE) system.

$$\gamma = 1+\frac{1}{\beta}+\frac{\tilde{\gamma}(1-\alpha)}{\sigma}\underbrace{\left(\frac{\tilde{\gamma}}{\alpha\beta}-\delta\right)}_{\frac{\bar{C}}{\bar{K}}} > 1+\frac{1}{\beta}$$

$$(1-V_{kk,1})\cdot(1-V_{kk,2}) = 1+V_{kk,1}\cdot V_{kk,2}-(V_{kk,1}+V_{kk,2}) = 1+\frac{1}{\beta}-\gamma < 0$$

$\Rightarrow 0 < V_{kk,1} < 1 < V_{kk,2} \Rightarrow V_{kk,2}$ is explosive root, unstable root should be deleted.

To see this, shut down shock $\hat{z}_t = 0 \Rightarrow \hat{k}_t = V_{kk}\hat{k}_{t-1}$, V_{kk} is the speed of convergence.

$|V_{kk}| < 1$: deviation in capital shrink over time $\Rightarrow$ economy goes to S.S.

$|V_{kk}| > 1$: deviation in capital grows over time $\Rightarrow$ economy explode, violates TVC.

From BK condition's angel, we have one pre-determined state $\hat{k}_{t-1}$, and one jump variable $\hat{c}_t$, We need # stable roots = # predetermined state $\Rightarrow$ Exactly 1 stable root $V_{kk,1}$

☆ We need to drop $V_{kk,2}$. Only $V_{kk,1}$ is stable root

Since $V_{kk} = \frac{\gamma}{2}-\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}} \Rightarrow V_{ck} = \frac{1}{\frac{\tilde{\gamma}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}\right)$

(ii) (2) and (4) are two equations about (V_{kz}, V_{cz}), solve for V_{cz} by substituting V_{kz} out.

$$\sigma V_{cz}(1-\psi) = \left[\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}\right)+\tilde{\delta}(1-\alpha)\right]\left[\frac{\tilde{\delta}}{\alpha\beta}-\left(\frac{\tilde{\delta}}{\alpha\beta}-\delta\right)V_{cz}\right]-\tilde{\delta}\psi$$

$$\Rightarrow V_{cz} = \frac{\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}\right)+\tilde{\delta}(1-\alpha)-\alpha\beta\psi}{\sigma(1-\psi)+\left[\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}\right)+\tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta}-\delta\right)} \times \frac{\tilde{\delta}}{\alpha\beta}.$$

$$\Rightarrow V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\tilde{\delta}}{\alpha\beta}\left(\frac{\tilde{\delta}}{\alpha\beta}-\delta\right)\cdot \frac{\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}\right)+\tilde{\delta}(1-\alpha)-\alpha\beta\psi}{\sigma(1-\psi)+\left[\sigma \frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta}-\frac{\gamma}{2}+\sqrt{\left(\frac{\gamma}{2}\right)^2-\frac{1}{\beta}}+\tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta}-\delta\right)}$$

Now we have:
$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t \\ \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t \end{cases}$$

Where, $V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}$, $V_{ck} = \frac{\bar{K}}{\bar{C}}\left(\frac{1}{\beta} - V_{kk}\right)$

$$V_{cz} = \frac{\sigma V_{ck} + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + [\sigma V_{ck} + \tilde{\delta}(1-\alpha)]\frac{\bar{C}}{\bar{K}}} \times \frac{\tilde{\delta}}{\alpha\beta}$$

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\bar{C}}{\bar{K}} V_{cz}.$$

Feed in the parameter values. quarterly model:

$\beta = 0.99$ match to 4.1% annual rate: $r_q = \frac{1}{\beta} - 1 \approx 1.01\%$ $r_a \approx (1+r_q)^4 - 1 \approx 4.1\%$

$\alpha = 0.36$ match to data capital income share

$\sigma = 1$ log utility. (usually $\sigma = 2$ implying $IES = \frac{1}{\sigma} = 0.5$)

$\delta = 0.025$ capital depreciate by 2.5% every quarter. (annual depreciate 10%)

$\bar{Z} = 1$ assume steady state tech level is normalized.

$\psi = 0.95$ persistence of productivity shock, how long shock in ε_t last

Then, we can get $(V_{kk}, V_{cz}, V_{kz}, V_{ck})$

6) Impulse response function (IRF)

Shock ε_t and then see how all variables respond.

eg: Trace out all variables for $\varepsilon_1 = 1$, $\varepsilon_t = 0$ for $t \geq 2$, start from S.S. (one-time temporary)

Since $\hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t$

$\Rightarrow \hat{z}_1 = \psi \hat{z}_0 + \varepsilon_1 = \varepsilon_1$. since we start from S.S. $\hat{z}_0 = 0$

recall hat $(\wedge)$ means deviation from S.S.

$\hat{z}_2 = \psi \hat{z}_1 + \varepsilon_2 = \psi \hat{z}_1 = \psi \varepsilon_1 \quad \Rightarrow \quad \hat{z}_t = \psi^{t-1} \varepsilon_1$ when $|\psi|<1$, shock ε_1 dies out as $t \to \infty$

....

& $\hat{k}_1 = V_{kk} \hat{k}_0 + V_{kz} \hat{z}_1 = V_{kz} \varepsilon_1$, at steady state $\hat{k}_0 = 0$

$\Rightarrow \hat{k}_2 = V_{kk} \hat{k}_1 + V_{kz} \hat{z}_2 = V_{kk} V_{kz} \varepsilon_1 + V_{kz} \psi \varepsilon_1 = (V_{kk} + \psi) V_{kz} \varepsilon_1$

$\dots \Rightarrow \hat{k}_t = V_{kk} \hat{k}_{t-1} + V_{kz} \psi^{t-1} \varepsilon_1 = \sum_{i=0}^{t-1} \left(\underbrace{V_{kk}^i}\, \psi^{t-i-1} \right) \underbrace{V_{kz}}\, \varepsilon_1$ → Constants

Every past TFP effect ψ^{t-i-1} gets stored/propagated through capital at rate V_{kk}^i

V_{kz} translates TFP effect to effect on capital

How to measure speed of convergence? Using half-life 半衰期

We have $\hat{k}_{t+h} = V_{kk}^h \hat{k}_t$, "half-life" is defined as

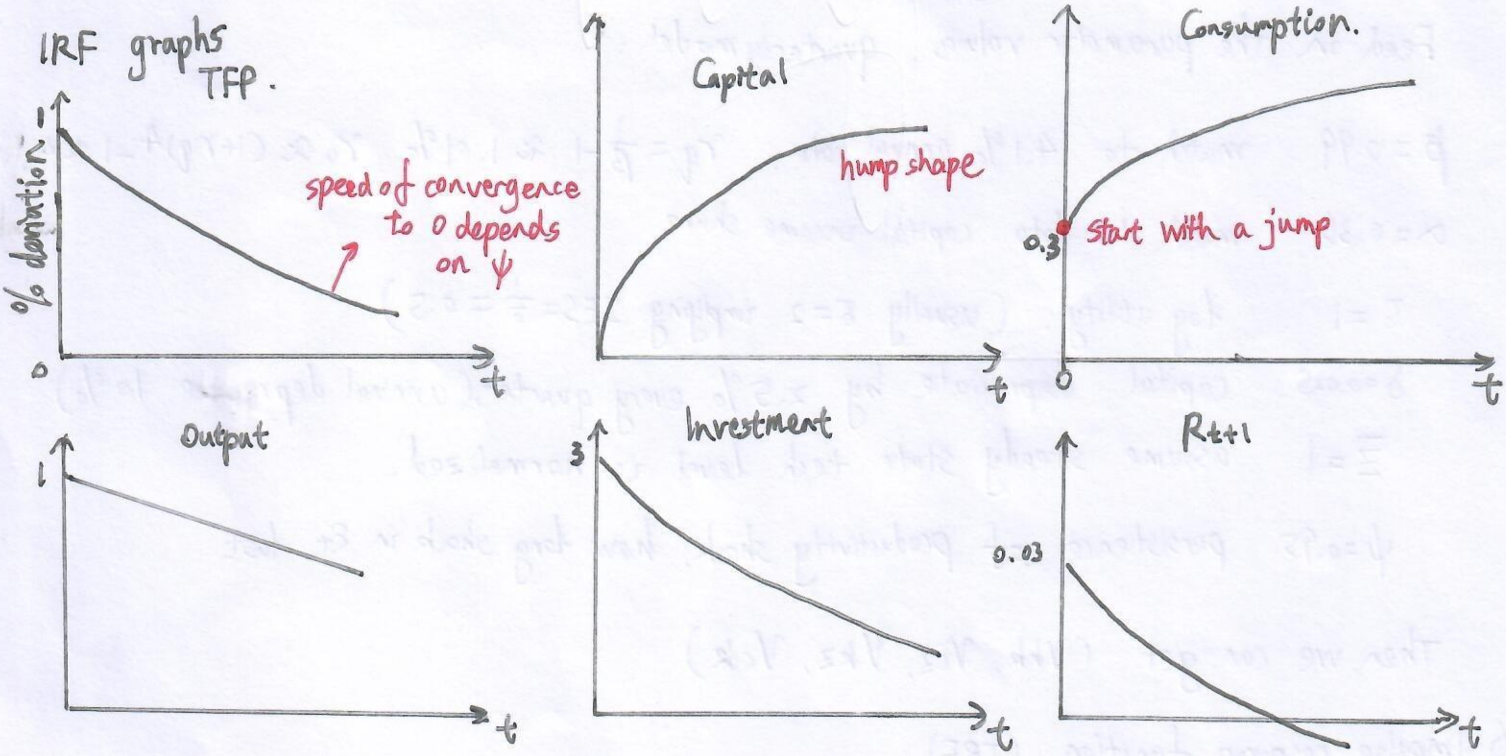
Capital accumulation half life: $\dfrac{\hat{k}_{t+h_{\frac{1}{2}}}}{\hat{k}_t} = \dfrac{1}{2} \Rightarrow V_{kk}^{h_{1/2}} = \dfrac{1}{2} \Rightarrow h_{1/2} = \dfrac{\ln(\frac{1}{2})}{\ln(V_{kk})}$ $(0 < V_{kk} < 1)$

TFP shock half life: $h_{1/2}^z = \dfrac{\ln(\frac{1}{2})}{\ln(\psi)}$

if $V_{kk} = 0.95$, $h^k_{1/2} \approx 13.5$ quarters ≈ 3.4 years

if $V_{kk} = 0.90$, $h^k_{1/2} \approx 6.6$ quarters ≈ 1.6 years

IRF graphs



Question: Why capital is hump shape?

$$\hat{k}_t = \sum_{i=0}^{t-1} \left(V_{kk}^i \psi^{t-i-1} \right) V_{kz} \varepsilon_1 = \begin{cases} V_{kz} \varepsilon_1 \dfrac{\psi^t - V_{kk}^t}{\psi - V_{kk}}, & \psi \neq V_{kk} \\ V_{kz} \varepsilon_1 \, t \cdot \psi^t, & \psi = V_{kk} \end{cases}$$

$\psi = V_{kk} \Rightarrow V_{kk}^{i}\, \psi^{t-i-1} = \psi^{i}\psi^{t-i-1} = \psi^{t-1}$

$\Rightarrow \hat{k}_t = \sum_{i=0}^{t-1} \psi^{t-1} V_{kz}\varepsilon_1 = t\cdot\psi^{t-1} V_{kz}\varepsilon_1$

$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{(t+1)\psi^t}{t\,\psi^{t-1}} = \psi(1+\frac{1}{t})$ — For small t, $1+\frac{1}{t}$ is big, ratio exceed 1, $\hat{k}_t$ keeps rising. For big t, $1+\frac{1}{t}\to 1$, ratio $\to \psi<1$, $\hat{k}_t$ falls.

peak happens at: $\psi(1+\frac{1}{t}) = 1 \Rightarrow t \approx \frac{\psi}{1-\psi}$

* Thus, $\psi = V_{kk}$, $\hat{k}_t$ rises at first, peak at $t\approx\frac{\psi}{1-\psi}$ and decays gradually

When $\psi \neq V_{kk}$: $\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{\psi^{t+1}-V_{kk}^{t+1}}{\psi^t - V_{kk}^t} \Rightarrow \frac{\hat{k}_2}{\hat{k}_1} = \psi + V_{kk}$ get an initial rise if $\psi+V_{kk}>1$

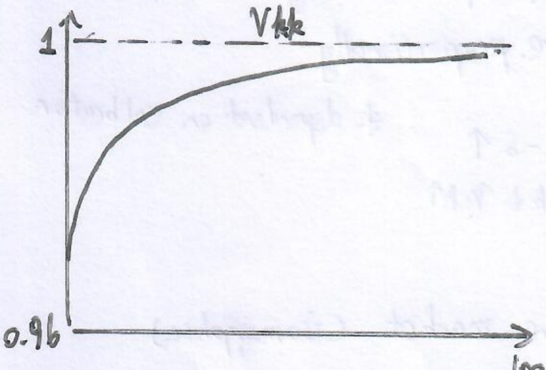
It peaks when $\hat{k}_{t+1} = \hat{k}_t \Rightarrow \psi^{t+1} - V_{kk}^{t+1} = \psi^t - V_{kk}^t \Rightarrow \psi^t(1-\psi) = V_{kk}^t(1-V_{kk}) \Rightarrow t^* = \frac{\ln\left(\frac{1-V_{kk}}{1-\psi}\right)}{\ln\left(\frac{\psi}{V_{kk}}\right)}$

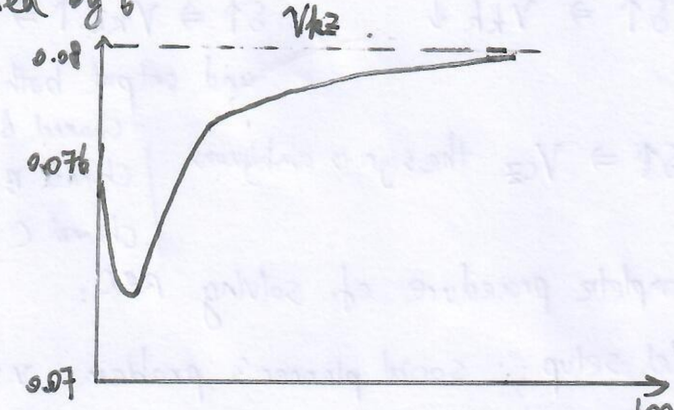
If $t^*>1$, IRF rises for a while then falls (hump), if $t^*\le 1$, peaks in first period and monotonically decay.

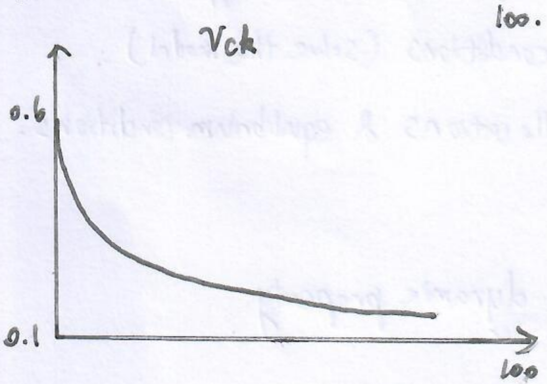
§ The effect of σ:

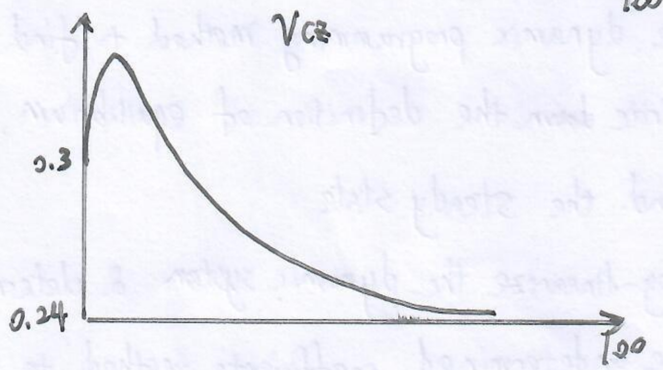
Supply only depends on patience β and δ, labor is fixed. Demand: investment covers depreciation. $\Rightarrow$ does not have long-run effect

But do have short-run effect: governed by σ









1° $\sigma\uparrow \Rightarrow V_{kk}\uparrow \to 1 \Rightarrow$ capital is more persistent $\Rightarrow$ slower convergence

* households hate shifting consumption across time (lower IES)

2° $\sigma\uparrow \Rightarrow$ IES $\downarrow \Rightarrow V_{ck}\downarrow$ marginal utility is more responsive to change in C. $\Rightarrow$ Consumption move less for any given incentive coming from the state k_{t-1}

3° $\sigma \uparrow \Rightarrow$ IES $\downarrow \Rightarrow V_{cz} \downarrow \Rightarrow$ consumption's direct response to the shock shrinks

4° $\sigma \uparrow \Rightarrow$ IES $\downarrow \Rightarrow V_{cz} \downarrow \Rightarrow V_{kz} \uparrow \Rightarrow$ After a TFP shock, consumption is reluctant to jump. more of the resources show up as investment/capital accumulation

Recall we solved the S.S. of the system:

$$\bar{C} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{K}$$

$$1 = \beta\,[\alpha \bar{Z}\bar{K}^{\alpha-1} + (1-\delta)]$$

Two variables $(\bar{K}, \bar{C})$ given parameters $(\bar{Z}, \beta, \alpha, \delta)$

$$\Rightarrow \begin{cases} \bar{K} = \left(\dfrac{\alpha \bar{Z}}{\frac{1}{\beta} - 1 + \delta}\right)^{\frac{1}{1-\alpha}} \\ \bar{Y} = \bar{Z}\bar{K}^{\alpha}, \quad \bar{C} = \bar{Y} - \delta \bar{K} \\ \dfrac{\bar{C}}{\bar{K}} = \dfrac{\bar{Y}}{\bar{K}} - \delta = \bar{Z}\bar{K}^{\alpha-1} - \delta = \dfrac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha} \end{cases}$$

5) The effect of δ (depreciation).

	$\bar{K}$	$\bar{Y}$	$\bar{C}$
$\delta = 0.025$	38	3.7	2.75
$\delta = 0.1$	6.4	1.95	1.3

Higher depreciation $\Rightarrow$ lower S.S. level

$\delta \uparrow \Rightarrow V_{kk} \downarrow$, $\delta \uparrow \Rightarrow V_{kz} \uparrow \Rightarrow$ less stock of capital and higher MPK, return and output both respond more proportionally.

$\delta \uparrow \Rightarrow V_{cz}$ the sign is ambiguous
$$\begin{cases} \text{Channel A:} & \delta \uparrow \Rightarrow \tilde{\delta} \uparrow \\ \text{Channel B:} & \delta \uparrow \Rightarrow \frac{\gamma}{\alpha\beta} - \delta \uparrow \\ \text{Channel C:} & \delta \uparrow \Rightarrow V_{kk} \downarrow \; V_{ck} \uparrow \end{cases}$$
* dependent on calibration

The complete procedure of solving RBC:

① Model setup: social planner's problem v.s. competitive market (isomorphic)

② Use dynamic programming method to find necessary conditions (solve the model).

③ Write down the definition of equilibrium, including allocations & equilibrium conditions.

④ Find the steady state

⑤ Log-linearize the dynamic system & determine local dynamic property.

⑥ Use undetermined coefficients method to solve LoM:
$$k_t = V_{kk} k_{t-1} + V_{kz} z_t$$
and then the other variables c_t

⑦ Calibrate and simulate the model.

§ State space approach: 2-D state space / saddle path analysis

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t \qquad \text{Let } \hat{z}_t = 0$$

$$\Rightarrow \quad \hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t$$

$$\Rightarrow \quad \hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1} - \frac{\bar{K}}{\bar{C}}\hat{k}_t$$

$$\sigma E_t\hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \hat{\delta}E_t\hat{z}_{t+1} = \sigma\hat{c}_t \qquad \text{Let } \hat{z}_t = 0.$$

$$\Rightarrow \quad \sigma(\hat{c}_t - \hat{c}_{t+1}) - [1-\beta(1-\delta)](1-\alpha)\hat{k}_t = 0.$$

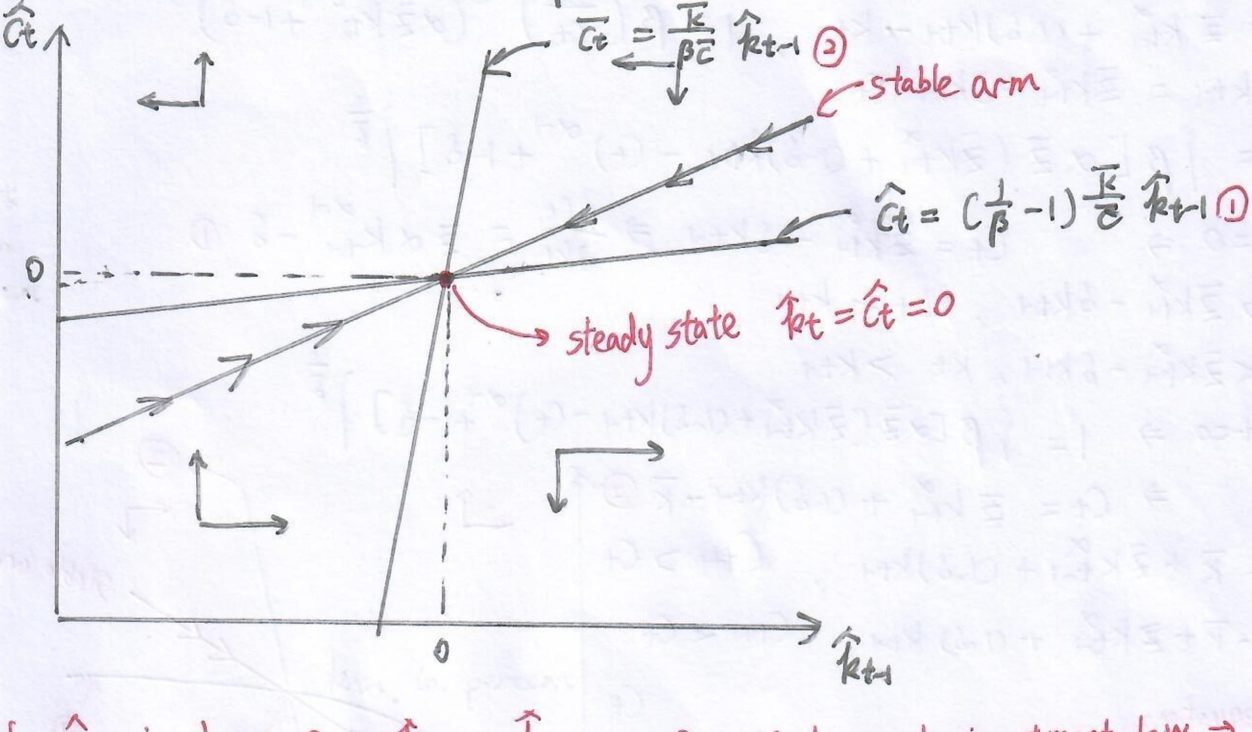
Rearrange:

$$\begin{cases} \hat{k}_t - \hat{k}_{t-1} = (\frac{1}{\beta}-1)\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t \\ \hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma}[1-\beta(1-\delta)](1-\alpha)\underbrace{(\frac{\bar{C}}{\bar{K}}\hat{c}_t - \frac{1}{\beta}\hat{k}_{t-1})}_{=-\hat{k}_t} \end{cases}$$

Steady state:

$$\hat{k}_t = \hat{k}_{t-1} \quad \Rightarrow \quad \hat{c}_t = (\frac{1}{\beta}-1)\frac{\bar{K}}{\bar{C}}\hat{k}_{t-1}$$

$$\hat{c}_{t+1} = \hat{c}_t \quad \Rightarrow \quad \hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1}$$



If $\hat{c}_t$ is above ①, $\hat{k}_t < \hat{k}_{t-1}$, consume too much, investment low ⇒ capital falls

If $\hat{c}_t$ is below ①, $\hat{k}_t > \hat{k}_{t-1}$, consume too little, save more ⇒ capital rises.

If $\hat{c}_t$ is above (left to) ②, $\hat{c}_{t+1} > \hat{c}_t$, consume more.

If $\hat{c}_t$ is below (right to) ②, $\hat{c}_{t+1} < \hat{c}_t$, consume less.

How to see these relations:

1° $\hat{c}_t$ above ① $\Rightarrow$ $\hat{c}_t > (\frac{1}{\beta}-1)\frac{\bar{k}}{\bar{c}}\hat{k}_{t-1}$ $\Rightarrow$ $\frac{\bar{c}}{\bar{k}}\hat{c}_t - (\frac{1}{\beta}-1)\hat{k}_{t-1} > 0$

$\Rightarrow$ $\underbrace{(\frac{1}{\beta}-1)\hat{k}_{t-1} - \hat{k}_t + \hat{k}_{t-1}}_{\frac{\bar{c}}{\bar{k}}\hat{c}_t} - (\frac{1}{\beta}-1)\hat{k}_{t-1} > 0 \Rightarrow \hat{k}_t < \hat{k}_{t-1}$

2° $\hat{c}_t$ above ② $\Rightarrow$ $\hat{c}_t > \frac{\bar{k}}{\beta\bar{c}}\hat{k}_{t-1}$ $\Rightarrow$ $\frac{\bar{c}}{\bar{k}}\hat{c}_t - \frac{1}{\beta}\hat{k}_{t-1} > 0$

$\hat{c}_{t+1} - \hat{c}_t = \underbrace{\frac{1}{\sigma}[1-\beta(1-\delta)](1-\alpha)}_{>0}\left(\frac{\bar{c}}{\bar{k}}\hat{c}_t - \frac{1}{\beta}\hat{k}_{t-1}\right)$ $\Rightarrow$ $\hat{c}_{t+1} - \hat{c}_t > 0$

$\Rightarrow$ $\hat{c}_{t+1} > \hat{c}_t$

* Stable arm is the unique saddle path that converges

$\hat{c}_t = \nu_{ck}\hat{k}_{t-1}$

speed of convergence: $\hat{k}_t = \nu_{kk}\hat{k}_{t-1} \Rightarrow \hat{k}_{t+h} = \nu_{kk}^{h}\hat{k}_t$

Nature gives $\hat{k}_{t-1}$, household choose $\hat{c}_t$ to land exactly on the stable arm then economy follows the arrows along that arm to S.S.

* off the stable arm, unstable and diverges

3) Redo in levels: Set $z_t = \bar{z}$

$c_t = \bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - k_t$, $1 = \beta\left(\frac{c_{t+1}}{c_t}\right)^{-\sigma}(\alpha\bar{z}k_t^{\alpha-1} + 1 - \delta)$

$$\Rightarrow \begin{cases} k_t - k_{t-1} = \bar{z}k_{t-1}^{\alpha} - \delta k_{t-1} - c_t \\ \frac{c_{t+1}}{c_t} = \left\{\beta\left[\alpha\bar{z}\left(\bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - c_t\right)^{\alpha-1} + 1 - \delta\right]\right\}^{\frac{1}{\sigma}} \end{cases}$$

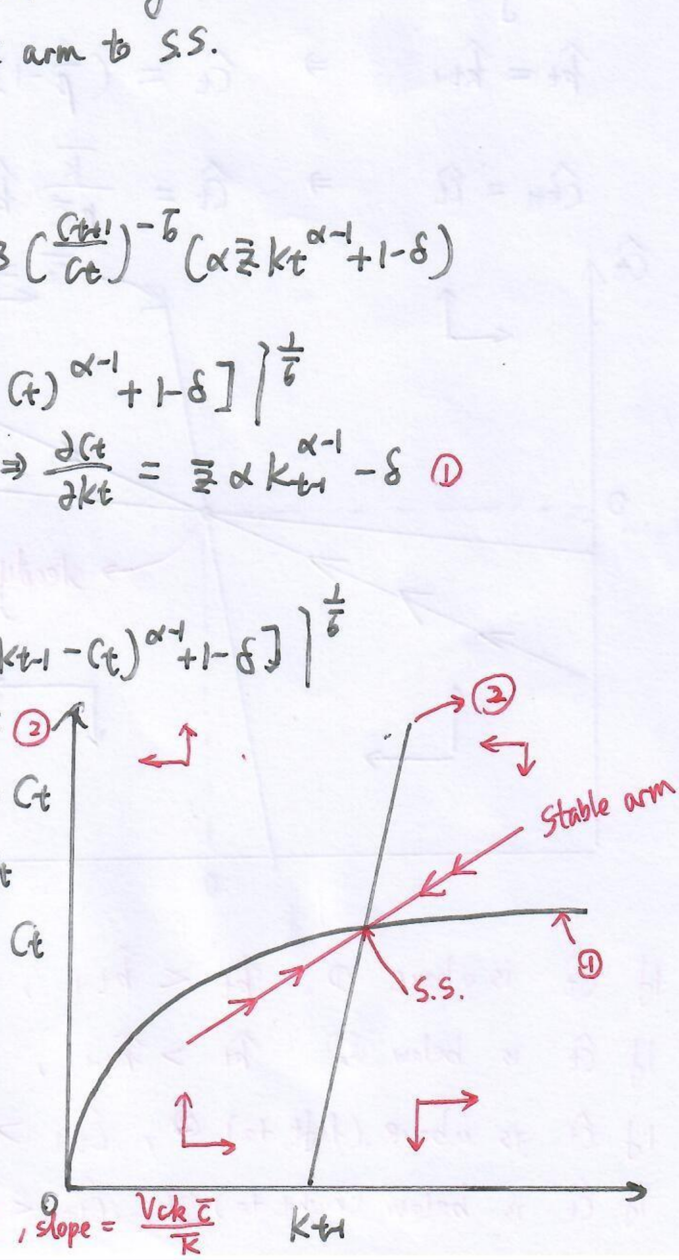
At S.S. $\Delta k_t = 0 \Rightarrow$ $c_t = \bar{z}k_{t-1}^{\alpha} - \delta k_{t-1} \Rightarrow \frac{\partial c_t}{\partial k_t} = \bar{z}\alpha k_{t-1}^{\alpha-1} - \delta$ ①

$$\begin{cases} \text{If } c_t > \bar{z}k_{t-1}^{\alpha} - \delta k_{t-1}, \ k_t < k_{t-1} \\ \text{If } c_t < \bar{z}k_{t-1}^{\alpha} - \delta k_{t-1}, \ k_t > k_{t-1} \end{cases}$$

At S.S. $\Delta c_t = 0 \Rightarrow 1 = \left\{\beta\left[\alpha\bar{z}\left(\bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - c_t\right)^{\alpha-1} + 1 - \delta\right]\right\}^{\frac{1}{\sigma}}$

$\Rightarrow c_t = \bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - \bar{k}$ ②

$$\begin{cases} \text{If } c_t > -\bar{k} + \bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1}, \ c_{t+1} > c_t \\ \text{If } c_t < -\bar{k} + \bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1}, \ c_{t+1} < c_t \end{cases}$$



Steady state equations.

$$\begin{cases} c_t = \bar{z}k_{t-1}^{\alpha} - \delta k_{t-1} & ① \\ c_t = \bar{z}k_{t-1}^{\alpha} + (1-\delta)k_{t-1} - \bar{k} & ② \end{cases}$$

Why stable arm still linear? $\hat{c}_t = \nu_{ck}\hat{k}_{t-1} \Rightarrow$

$\frac{c_t - \bar{c}}{\bar{c}} = \nu_{ck}\frac{k_t - \bar{k}}{\bar{k}} \Rightarrow c_t = \frac{\nu_{ck}\bar{c}}{\bar{k}}k_t - \nu_{ck}\bar{c} + \bar{c}$, slope $= \frac{\nu_{ck}\bar{c}}{\bar{k}}$